Contributed Articles

Compact Simplex Method

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ABSTRACT

The following documented algorithm solves the standard linear programming problem of optimizing a linear form subject to linear inequality or equality constraints and nonnegativity conditions. The solution procedure incorporates the standard simplex method with no embellishments. There is only one loop corresponding to the basic simplex iteration.

The algorithm initializes by making all constraint "right-hand-sides" nonnegative, forming augmented constraint matrix M and criterion vector Z by adding slacks, artificials, and right-hand-sides, and forming the basis of column B. The main simplex loop then changes the basis iteratively to improve the objective function until an optimal solution is obtained or iteration is terminated by virtue of no solution existent, an unbounded solution, or a predefined limit. The algorithm also iterates to delineate alternate optimal solutions, if any, again subject to a predefined limit. If optimal solutions are obtained, the rows of the matrices PRIMAL and DUAL contain the corresponding primal and dual solutions. The scalar OBJ yields the optimal value of the objective function.

For problems of around fifty constraints, this algorithm generally runs in less than ten CPU seconds on an IBM 3081 under VM/CMS, depending on the number of variables and the configuration of numbers and signs. For problems of one-hundred constraints or more, it is slow and is probably an inappropriate solution vehicle, although it will run on a big machine with an expanded workspace. One test problem with 111 constraints, 182 variables, and mixed signs ran in a little under one CPU minute, but this was much slower than more refined linear programming codes.

```
∇      K←SIMPLEX U ;NR;I;I;J;NA;NS;NC;J;NT;V;X;L;N;NO;⎕IO
[1]    ⍝ THIS FUNCTION SOLVES THE FOLLOWING LINEAR PROGRAM VIA THE
[2]    ⍝ BASIC SIMPLEX METHOD: DETERMINE ONE OR MORE VECTORS X
[3]    ⍝ WHICH OPTIMIZE THE OBJECTIVE FUNCTION C+.×X SUBJECT TO THE
[4]    ⍝ CONSTRAINTS X≥0: (((S='≤')+.×X)≤(S='≤')/B; ((S='≥')≠A)
[5]    ⍝ +.×X)≥(S='≥')/B; AND (((S='=')≠A)+.×X)=(S='=')/B.  THE
[6]    ⍝ ARGUMENT U IS EQUAL TO ONE FOR MAXIMIZATION AND MINUS ONE
[7]    ⍝ FOR MINIMIZATION.  A, B, AND C ARE NUMERIC GLOBAL INPUTS
[8]    ⍝ WITH (⍴A)=(⍴B),⍴C.  S IS A CHARACTER GLOBAL INPUT VECTOR
[9]    ⍝ OF SIGNS '≤≥=' WITH (⍴S)=⍴B.  OTHER NUMERIC GLOBAL INPUT
[10]   ⍝ SCALARS ARE BIGM, WHICH IS A POSITIVE NUMBER SIGNIFICANTLY
[11]   ⍝ LARGER THAN ⌈/|C USED TO FORCE ARTIFICIAL VARIABLES OUT OF
[12]   ⍝ THE BASIS, MAITER, WHICH LIMITS THE NUMBER OF SIMPLEX
[13]   ⍝ ITERATIONS, MAXOPT, WHICH LIMITS THE NUMBER OF ALTERNATIVE
[14]   ⍝ OPTIMAL SOLUTIONS, AND PR, A NUMBER LIKE 10 OR 12 WHICH
[15]   ⍝ GOVERNS ROUNDING TO AVOID BOGUS REPRESENTATIONS OF ZERO.
[16]   ⍝ ROUNDING IS PERFORMED BY THE UTILITY SUBFUNCTION RND. THE
[17]   ⍝ EXPLICIT RESULT K IS EQUAL TO( ZERO IF AN OPTIMAL SOLUTION
[18]   ⍝ IS DETERMINED; ONE IF NO SOLUTION EXISTS; TWO IF THE SOLUTION
[19]   ⍝ IS UNBOUNDED; AND THREE IF THE ITERATION LIMIT IS ACTIVE. IF
[20]   ⍝ AN OPTIMUM SOLUTION IS DETERMINED, THE ROWS OF PRIMAL CONTAIN
[21]   ⍝ ONE OR MORE DISTINCT ALTERNATIVE PRIMAL SOLUTIONS, THE ROWS OF
[22]   ⍝ DUAL CONTAIN CORRESPONDING (NOT NECESSARILY DISTINCT) DUAL
[23]   ⍝ SOLUTIONS, AND OBJ CONTAINS THE OPTIMAL VALUE OF THE OBJECTIVE
[24]   ⍝ FUNCTION.
```

```
[25]   ⍝ REF: G. HADLEY, LINEAR PROGRAMMING, ADDISON-WESLEY, 1962.
[26]   ⎕IO←1
[27]   I←(⍳NR)∘.=⍳NR←⍴B
[28]   B[I]←-B[I←(B<0)/⍳NR]
[29]   A[I;]←-A[I;]
[30]   S[J,K]←((⍴J←(S[I]='≥')/I)⍴'≤'),(⍴K←(S[I]='≤')/I)⍴'≥'
[31]   M←A,((S='≤')/I),(-(S='≥')/I),((S≠'≤')/I),B
[32]   Z0←(U×C),((NS←+/S≠'=')⍴0),((NA←+/S≠'≤')⍴-BIGM),0
[33]   Z←PR RND-Z0+BIGM×+⌿(S≠'≤')⌿M
[34]   B←(NC+⍳+/S='≤'),(NC←⍴C)+NS+⍳NA
[35]   B←B[J←⍋((S='≤')/⍳NR),((S='≥')/⍳NR),(S='=')/⍳NR]
[36]   PRIMAL←(0,NC)⍴0
[37]   DUAL←(0,NR)⍴0
[38]   NT←NC+NS+NA+N←NO←K←1
[39]  L1: →(0>V←⌊/¯1↓Z←(PR-2)RND Z[N←(~(⍳NT)∈B)/⍳NT])/L3
[40]   →(∨/0<,M[(B∈(-NA)↓⍳NT-1)/⍳NR;NT])/0,X←(I←B≤NC)/,M[;NT]
[41]   →((V=0)∧1=∨/(X←((⍳NC)∈I)\X[⍋I←I/B])∧.=⍉PRIMAL)/L2
[42]   PRIMAL←PRIMAL,[1]X
[43]   I←(NC+⍳NS),NC←NS+(+/S='≥')+⍳+/S='='
[44]   DUAL←DUAL,[1]PR RND((I∈N)\(Z+Z0)[(N∈I)/N])[J]
[45]  L2: →((V>0)∨MAXQPT<NO←NO+1)/(K←0),OBJ←U×¯1↑Z
[46]  L3: L←0<⌈⌿(M[;N])[;J←(V=¯1↑Z)/⍳NT-NR+1]
[47]   →((0>V),(∨/L),1)/L4,L5,0
[48]  L4: →(∧/L)/L5
[49]   →0,K←2
[50]  L5: J←J[?⍴J←L/J]
[51]   V←⌊/X←(,M[L;NT])+,M[L←(0<,M[;N[J]])/⍳NR;N[J]]
[52]   B[I←I[?⍴I←(V=X)/L]]←N[J]
[53]   M←PR RND M-(,M[;N[J]])∘.×V←(,M[I;])+,M[I;N[J]]
[54]   Z←PR RND Z-Z[N[J]]×,M[I;]←PR RND V
[55]   →(MAXITER≥N←N+1)/L1
[56]   →0,K←3
       ∇

       ∇      Z←X RND Y
[1]    Z←(10*-X)×⌊.5+(10*X)×Y
       ∇
```

Example 1. Production Planning

Suppose we produce five products in three departments subject to the following constraints, for a week say.

Product	Unit Margin	Dept. #1	Dept. #2	Dept. #3	Minimum Units
1	10	6	4	2	50
2	10	10	8	3	30
3	30	20	10	4	20
4	15	10	6	2	10
5	5	5	3	1	20
Capacity	(Man-Hours)	2,000	1,200	500	

We assume that there is no limit on demand for any product but that the firm must decide on its production mix to maximize profit subject to capacity constraints on man-hours and also to minimum demand requirements from key customers. Our algorithm solves this problem as follows:

```
      BIGM
1000

      PR
12

      MAXITER
300

      MAXOPT
100

      A
 6  10  20  10  5
 4   8  10   6  3
 2   3   4   2  1
 1   0   0   0  0
 0   1   0   0  0
 0   0   1   0  0
 0   0   0   1  0
 0   0   0   0  1

      B
2000  1200  500  50  30  20  10  20

      C
10  20  30  15  5

      S
≤≤≤≥≥≥≥≥

      SIMPLEX 1
0

      2▼OBJ
3216.67

      2▼PRIMAL
50.00  43.33  53.33  10.00  20.00

      2▼DUAL
.67  1.67  .00  .67  .00  .00  1.67  3.33
```

This problem has a unique solution which produces the minimum requirements for products 1, 4 and 5 and more of the profitable products and 2 and 3. For problems like this with inequality constraints, the dual solution yields a so-called "shadow-price" interpretation, e.g., for a small increase in capacity of department #1, say to 2001, the optimal objective function increases by .67 or for a small decrease in the minimum requirement for product 5, say to 19, the optimal objective function increases by 3.33.

Example 2. Transportation Problem

Suppose we ship a product from four plants to five stores subject to the following constraints, again for a week say.

Plant/Store	1	2	3	4	5	Unit Supply
1	10	15	20	25	999	25
2	10	15	20	999	15	50
3	15	20	999	15	20	100
4	25	999	20	15	20	25
Unit demand	25	100	25	25	25	

The objective is to satisfy demand at minimum cost. Note that total demand equals total supply and the 999s indicate infeasible routings. Although the standard simplex method is not the most efficient way to solve a transportation problem, our algorithm solves this problem as follows:

```
      A
1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1
1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0
0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0
0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0
0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0
```

```
      B
25 50 100 25 25 100 25 25

      C
10 15 20 25 999 10 15 20 999 15 15 20 999 15 20 25 999 20 15 20

      S
========

      SIMPLEX¯1
0

      OBJ
3375

      PRIMAL
25  0 0 0 0  0 25 0 0 25  0 75 0 25  0 0 0 25 0 0
 0 25 0 0 0  0 25 0 0 25 25 50 0 25  0 0 0 25 0 0
 0 25 0 0 0 25  0 0 0 25  0 75 0 25  0 0 0 25 0 0
 0 25 0 0 0 25 25 0 0  0  0 50 0 25 25 0 0 25 0 0
25  0 0 0 0  0 50 0 0  0  0 50 0 25 25 0 0 25 0 0
 0 25 0 0 0  0 50 0 0  0 25 25 0 25 25 0 0 25 0 0

      0▼DUAL
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
¯15 ¯15 ¯20 ¯15 5 0 ¯5 5
```

Note that we have vectorized input parameters for solution, and we have also dropped a redundant constraint. This problem has six alternative optimal solutions which can be shaped for intuitive display as 6 4 5 ⍴ PRIMAL. The dual solution has no particular interpretation here.

Reference

[1] Hadley, G., *Linear Programming.* Addison-Wesley, 1962.

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Errata

Corrections to Compact Simplex Method

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On page 9, line 12 instead of "column B", read "column indices B".

These three comment lines come from the function "SIMPLEX":

```
[4]    ⍝ CONSTRAINTS X≥0: (((S='≤')/A)+.×X)≤(S='≤')/B; (((S='≥')/A)
[12]   ⍝ THE BASIS, MAXITER, WHICH LIMITS THE NUMBER OF SIMPLEX
[17]   ⍝ EXPLICIT RESULT K IS EQUAL TO: ZERO IF AN OPTIMAL SOLUTION
```

The table on p. 10 in Example 1, "Production Planning" should be:

	Unit	Man-Hours-Unit			Minimum
Product	Margin	Dept. #1	Dept. #2	Dept. #3	Units
1	10	6	4	2	50
2	20	10	8	3	30
3	30	20	10	4	20
4	15	10	6	2	10
5	5	5	3	1	20
Capacity	(Man-Hours)	2,000	1,200	500	

The table on p. 11 in Example 2, "Transportation Planning" should be:

	Unit Shipping Costs					Unit
Plant/Store	1	2	3	4	5	Supply
1	10	15	20	25	999	25
2	10	15	20	999	15	50
3	15	20	999	15	20	100
4	25	999	20	15	20	25
Unit demand	25	100	25	25	25	

Correction to **APL and Elementary Logic**

by Nollaig MacKenzie

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On page 21, column 2, line 30 instead of `'(',S1,'∧≤',S2`, read `'(',S1,')≤',S2`.